

Why Early COVID-19 Resists Automated Detection: A Radiological Audit of Physics-Based Feature Discriminability

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Abstract. Early-stage COVID-19 CT classification is fundamentally hard: mild Ground-Glass Opacities (GGOs) are too focal and heterogeneous to produce consistent whole-lung radiological signatures, leaving both interpretable and black-box AI systems at a performance ceiling that has not been rigorously explained. We present a controlled radiological audit using the Physics Attribute-Regularized Variational Autoencoder (PAR-VAE), a generative model constrained to encode 14 clinically interpretable radiological attributes grounded in X-ray attenuation physics. Across $\sim 5,000$ CT slices from MosMedData, we confirm an 84% class overlap in the global physics feature space, and show this ceiling is model-independent: a CNN baseline achieves only a narrow 2.4 AUC point advantage on mild disease (69.8% vs 67.4%). What separates the models is not mild-disease accuracy but clinical safety under severity shift: on severe COVID-19 (CT-3), PAR-VAE achieves $99.3 \pm 1.0\%$ AUC and misses only 1.6% of cases, while the CNN achieves only $66.0 \pm 7.5\%$ AUC and misses $46.9 \pm 19.7\%$ of all severe COVID-19 patients across seeds. Physics constraints act as implicit regularization, preventing the shortcut learning that causes CNN to memorize training noise rather than genuine pathology. A patch-level ablation further confirms the ceiling is biological: finer spatial granularity increases class overlap, because GGOs are too focal to produce consistent statistics even within individual patches. A cross-scanner transfer experiment additionally shows that PAR-VAE makes domain shift visible and quantifiable (R^2 drops from 0.972 to 0.320 across scanners, $\Delta \text{mean HU} = 482$ units), while CNN degrades silently. These findings help us observe the interpretability-performance tradeoff in the early COVID-19 detection which is mainly a consequence of biology and not modeling choice, and physics-aligned models offer a deployment safe path compared to Black box models.

Keywords: COVID-19 CT · Ground-Glass Opacity · Interpretable AI · Physics-Informed Machine Learning · Variational Autoencoder · Clinical Decision Support

1 Introduction

This work examines the discriminative limit of clinically interpretable radiological features across COVID-19 severity levels in a 2D whole-lung setting.

Early COVID-19 manifests primarily as Ground-Glass Opacities: focal, hazy regions of increased lung density caused by partial alveolar filling with inflammatory exudate. In mild disease (less than 25% lung involvement), GGOs typically occupy a small, peripheral portion of the parenchyma. Automated systems computing global statistics such as mean lung density, texture entropy, gradient magnitude across the full lung field effectively average the focal GGO signal over surrounding normal tissue, diluting it toward invisibility. This is not a failure of feature engineering. It is a consequence of lesion biology, and no amount of architectural sophistication changes this if the input representation remains global.

The clinical significance is considerable. High-accuracy CNNs appear to solve the detection problem but accumulating evidence shows they exploit non-pathological shortcuts like scanner artifacts, patient positioning biases and dataset-specific correlations rather than genuine radiological findings [[2]]. A model that learns to identify the scanner rather than the disease fails silently when deployed across hospital systems, with no internal signal that anything is wrong. Our CNN baseline provides direct evidence of this in the present work: it achieves 98% training accuracy on mild disease but the best validation performance occurs at epoch 3–17 of 50, indicating rapid memorization of training noise rather than pathological learning. Physics based approaches avoid this by aligning the representation to verifiable radiological measurements.

Our contributions are:

- A rigorous radiological feature extraction pipeline on $\sim 5,000$ CT slices with a 9-stage data integrity protocol and volume-level patient splitting, producing an audit of global feature discriminability across three disease severity levels.
- Model-independent confirmation that the $\approx 67\%$ AUC ceiling on mild COVID-19 is biological: both PAR-VAE and a CNN baseline hit the same wall, while PAR-VAE maintains near-perfect performance on severe disease where CNN catastrophically fails.
- A patch-level ablation showing finer spatial granularity increases class overlap, providing a mechanistic, biologically grounded explanation for the ceiling.
- A cross-scanner transfer experiment demonstrating that PAR-VAE makes domain shift visible and quantifiable, while CNN degrades silently: a clinically important interpretability advantage.

2 Related Work

Automated COVID-19 Detection from CT. Deep learning approaches achieve high accuracy on COVID-19 CT classification [7] but systematically exploit non-pathological shortcuts rather than genuine radiological findings [2], raising deployment safety concerns across hospital systems. Our work directly quantifies the cost of this shortcut learning: despite a narrow CT-1 advantage, CNN misses 46.9% of severe COVID-19 cases on average across seeds, while PAR-VAE misses 1.6%.

Radiomics and Physics-Based Features. Radiomic analysis [8] extracts quantitative imaging features from segmented regions to characterize tissue properties. Wang et al. [17] showed that radiomic features from expert-annotated lesion regions achieve strong performance, but manual annotation limits clinical scalability. We use the same class of features extracted automatically from whole-lung segmentation not as direct classifiers, but as supervisory constraints on a generative model’s latent space, allowing us to audit the information content of a physics-aligned representation without annotation.

Interpretable Medical AI. Physics-Informed Machine Learning [6] constrains neural representations with domain-validated knowledge. Concept Bottleneck Models [5] provide interpretability through clinician-defined concepts but require inference-time annotation. Attribute-constrained VAEs [1] align latent dimensions with semantic attributes for disentanglement; our PAR-VAE uses established clinical measurements specifically to audit discriminative capacity which is a different scientific goal. Our approach constrains the latent space during training only, requiring no inference-time annotation while producing clinically grounded representations that generalize more robustly than unconstrained models.

Domain Shift and Hidden Stratification. Oakden-Rayner et al. [11] demonstrated that models frequently fail systematically on important patient subgroups, concealed by aggregate performance metrics. Our transfer learning experiment provides a concrete instance: the R^2 drop from 0.972 to 0.320 when moving from MosMedData to COVID-CT-MD quantifies a scanner calibration gap that CNN cannot detect or communicate, while PAR-VAE makes it explicit and diagnostic.

3 Radiological Study Design

The central question is whether globally computed interpretable radiological features carry sufficient discriminative signal for clinical decision support. All methodological choices follow from this framing.

3.1 Dataset and Cohort Construction

We use MosMedData [10], a publicly available cohort of 1,110 patients (42% male, 56% female, median age 47 years) with five-level CT severity stratification from CT-0 (no pneumonia) to CT-4 (critical). This stratification is central: it allows disease severity to function as a controlled variable, enabling direct measurement of how global feature discriminability changes with GGO burden. All patient volumes were split at the volume level (70/15/15 train/val/test) to prevent a model from recognizing patient anatomy rather than pathology; Chi-square testing confirmed label balance across splits ($p = 0.521$). Further data integrity checks are detailed in the Supplementary Material.

We constructed three binary classification tasks representing clinically distinct diagnostic challenges:

- *Cohort A (CT-1 vs. CT-0)*: Mild COVID-19 versus normal lung tissue. GGOs cover less than 25% of the lung which is the hard early detection case with 5,500 balanced slices (2,750 per class).
- *Cohort B (CT-2 vs. CT-0)*: Moderate disease (25–50% involvement), testing how discriminability improves with severity. 1,700 slices; the smaller CT-2 natural prevalence (125 volumes vs. 684 for CT-1) means CT-2 conclusions are directional rather than definitive.
- *Cohort C (CT-3 vs. CT-0)*: Severe disease (50–75%), testing the upper bound of global feature discriminability and the critical clinical scenario where missed cases carry the highest risk.

3.2 Radiological Feature Engineering

We engineered a 14-dimensional attribute vector $y_{\text{phys}} \in \mathbb{R}^{14}$ grounded in X-ray attenuation physics. The Hounsfield Unit scale encodes tissue attenuation:

$$HU = 1000 \times \frac{\mu_{\text{tissue}} - \mu_{\text{water}}}{\mu_{\text{water}} - \mu_{\text{air}}} \quad (1)$$

GGO formation shifts HU from air-like values (≈ -900 HU) toward the fluid range (-500 to -300 HU) [16]. All 14 features are computed within the automatically segmented lung mask, spanning four categories:

- *Tissue density (7 features)*: Mean HU, standard deviation, and five percentiles (10th–90th) capturing the overall density shift from GGO accumulation.
- *Boundary sharpness (2 features)*: Sobel gradient mean and standard deviation: GGOs blur vessel boundaries, reducing gradient magnitude [15], one of the primary visual cues radiologists use.
- *Lung geometry (2 features)*: Mask area and fractional occupancy: atelectasis and reduced inflation from inflammatory disease alter effective lung volume [14].
- *Texture (3 features)*: GLCM contrast, homogeneity, and entropy: GGOs produce “crazy-paving” patterns with increased spatial heterogeneity and reduced homogeneity [9].

Lung masks were generated by threshold-based morphological segmentation ($[-900, 100]$ HU, 5×5 kernels for closing and opening) without a pre-trained segmentation model, a deliberate choice to avoid introducing a second set of learned representations that could confound the discriminability audit. Up to 30 equidistant slices were extracted per volume from the central quartiles $[[0.25S], [0.75S]]$, focusing on parenchyma where GGOs are most prevalent and avoiding non-informative apical and basal slices. All features were standardized with `StandardScaler` prior to training.

4 PAR-VAE: A Generative Auditing Instrument

4.1 Motivation and Architecture

A standard classifier trained on radiological features reveals whether those features are predictive, but not *how much discriminative information they contain* or *why they fail*. The PAR-VAE addresses this by augmenting a Variational Autoencoder [4] with a physics attribute predictor that regularizes the latent space. By forcing the encoder to organize its internal representation around the same 14 measurements a radiologist would use to describe a scan, PAR-VAE makes the discriminative ceiling of interpretable representations both measurable and explainable.

The architecture has three components: an *encoder* (5-layer CNN mapping 512×512 CT slices to an 85-dimensional latent space, Leaky-ReLU activations, Batch Normalization); a *decoder* (mirrored 5-layer CNN, Tanh output); and a *physics attribute predictor* (3-layer MLP predicting the 14 radiological attributes from latent means μ_z). The 85-dimensional space is implicitly partitioned: 14 dimensions are regularized toward clinical attributes, 71 remain free for residual information not explained by global physics. Comparing classifiers trained on each subset directly reveals how much discriminative information resides in physics-aligned versus unconstrained portions of the representation.

The CNN baseline uses the same encoder architecture without the attribute predictor or physics constraints, trained with cross-entropy loss on class labels directly. This provides a model-independent comparison: if PAR-VAE and CNN hit the same AUC ceiling on mild disease, the ceiling is not architectural but biological.

4.2 Training Strategy

The training objective balances reconstruction, latent regularization, and physics alignment:

$$\mathcal{L} = \mathcal{L}_{\text{recon}} + \beta \cdot \mathcal{L}_{KL} + \lambda \cdot \mathcal{L}_{\text{attr}} \quad (2)$$

where $\mathcal{L}_{\text{recon}} = \text{MSE}(x, \hat{x})$, $\mathcal{L}_{KL} = D_{KL}(q(z|x) \parallel \mathcal{N}(0, I))$, and $\mathcal{L}_{\text{attr}} = \text{MSE}(a, \hat{a})$. Raising β too early causes posterior collapse (the encoder degenerates to the prior, $KL < 5$); raising λ too early locks in attributes before reconstruction stabilizes. Our 3-phase annealing curriculum (Table 1) addresses both failure modes simultaneously.

We use AdamW with learning rates 1×10^{-4} (encoder/decoder) and 5×10^{-4} (predictor), cosine annealing, gradient clipping at 1.0.

5 Results

5.1 Physics Alignment

PAR-VAE achieves mean $R^2 = 0.969$ (validation) and 0.972 (test) across all 14 radiological attributes (Table 2). The near-zero val-test gap of 0.003 confirms

Table 1. Three-phase annealing schedule preventing posterior collapse and attribute lock-in. Ablation evidence in Section 6.

Phase	Epochs	β/λ	Rationale
Physics-First	0–20	$10^{-4} \rightarrow 2 \cdot 10^{-4}$ / 1.5	Low KL allows latent to organise around clinical attributes first.
Gradual Balance	20–40	$2 \cdot 10^{-4} \rightarrow 5 \cdot 10^{-4}$ / $1.5 \rightarrow 3.0$	Physics supervision tightens as reconstruction stabilises.
Fine-Tune	40–50	$5 \cdot 10^{-4}$ / 3.0	Physics alignment maximised with stable latent structure.

stable generalization without overfitting. Boundary sharpness and lung geometry features achieve the highest alignment ($R^2 > 0.984$), reflecting their dependence on anatomical structure rather than subtle pathological shifts.

5.2 The Severity Gradient and CNN Baseline

Table 3 reports classification performance across all three severity levels for both PAR-VAE and the CNN baseline. Reading across the severity axis first: for mild disease (CT-1), CNN leads narrowly: $69.8 \pm 1.1\%$ test AUC versus PAR-VAE $67.4 \pm 1.4\%$. This 2.4 AUC point margin comes with a revealing cost: CNN achieves 99% training accuracy but its best validation epoch is 4–7 of 50, indicating rapid memorization of training noise rather than generalized pathological learning. PAR-VAE, with physics constraints acting as implicit regularization, converges more stably across seeds. Its variance (1.4%) is comparable to CNN’s (1.1%) at CT-1, but this stability advantage becomes decisive at CT-3.

The severity gradient tells the most important clinical story. Moving from CT-1 to CT-3, PAR-VAE AUC rises from 67.4% to $99.3 \pm 1.0\%$; CNN mean AUC remains at $66.0 \pm 7.5\%$, showing no meaningful benefit from increasing disease severity despite its CT-1 advantage. As GGO burden increases beyond 50% (CT-3), whole-lung statistics shift substantially enough that a physics-aligned model achieves near-perfect discrimination but the CNN, having learned shortcuts rather than pathological signals, fails to benefit from this shift and remains near its CT-1 ceiling. This difference reflects what each model has learned rather than a coincidence of hyperparameters.

On the CT-3 test set (183 severe COVID-19 patients), PAR-VAE misses only 3 cases (1.6% miss rate). CNN misses an average of 68 cases per seed, a mean $46.9 \pm 19.7\%$ miss rate ranging from 27.6% to 66.9% across seeds. Across seeds, CNN’s validation AUC at CT-3 0.714 ± 0.003 . Its test AUC, however, swings from 74.6% to 60.3%, showing a standard deviation of 7.5%, 26 times larger than the validation variance. The val-test gap widens across seeds: 2.8pp (seed 16), 8.0pp (seed 42), and 11.1pp (seed 999). This pattern is characteristic of a model that is overfit on training data and memorizes instead of learning features.

Table 2. Physics alignment performance measured by R^2 across cohorts (CT-1, CT-2, CT-3). All 14 features demonstrate strong alignment, indicating that the three-phase curriculum effectively encodes clinically interpretable attributes within the latent space.

Category	Feature	CT-1	CT-2	CT-3
Tissue Density	Mean HU	0.864	0.910	0.976
	HU Std. Dev.	0.832	0.927	0.981
	HU (10th percentile)	0.364	0.569	0.963
	HU (25th percentile)	0.794	0.846	0.956
	HU (50th percentile)	0.704	0.671	0.943
	HU (75th percentile)	0.814	0.803	0.962
	HU (90th percentile)	0.770	0.773	0.962
Lung Geometry	Mask Area	0.902	0.922	0.981
	Fractional Occupancy	0.909	0.927	0.984
Boundary Sharpness	Gradient Mean	0.791	0.820	0.989
	Gradient Std. Dev.	0.890	0.899	0.972
Texture	GLCM Contrast	0.819	0.805	0.979
	GLCM Homogeneity	0.843	0.880	0.982
	Texture Entropy	0.875	0.915	0.980
Mean		0.798	0.833	0.972

PAR-VAE does not exhibit this pattern. Its physics constraints force the latent representation to encode clinically grounded radiological attributes like tissue density distributions, boundary sharpness and texture that are meaningful regardless of which specific patients appear in the training fold. On CT-3, where GGO burden is high enough that these global attributes become genuinely discriminative, physics grounding translates directly into stable generalization: test AUC $99.3 \pm 1.0\%$ across seeds, with the val-test gap never exceeding 0.7pp. The small CT-3 cohort (45 volumes) that destabilizes CNN is handled robustly by PAR-VAE precisely because it does not rely on dataset-specific spatial shortcuts to make its predictions. This seed-to-seed instability of the CNN is itself a clinical risk: a deploying institution cannot know which version of CNN they are running. A system missing between 28% and 67% of severe cases represents a serious patient safety failure. PAR-VAE’s physics grounding, which appears to “cost” performance on mild disease, instead reveals that what CNN gains on CT-1 is noise memorization that actively harms severe-disease detection.

5.3 Class Overlap and Representational Analysis

Table 4 reports the overlap coefficient and Cohen’s d for physics-based and learned features. The 84% mean overlap confirms a fundamental biological ceiling. Critically, physics-based and learned features show nearly identical overlap statistics, showing that additional unconstrained latent dimensions recombine existing physics information rather than accessing qualitatively different dis-

Table 3. Classification performance: PAR-VAE vs. CNN baseline across severity levels (mean $\pm$ std over 3 seeds for all cohorts). **Bold** marks best test AUC per task. CNN leads narrowly on CT-1 (69.8% vs 67.4%); PAR-VAE leads decisively on CT-2 and CT-3 (99.3% vs 66.0% AUC).

Task	Model	Val Acc	Test Acc	Test F1	Test AUC
CT-1 vs CT-0 (Mild)	PAR-VAE (LogReg)	62.0 ± 2.1	62.6 ± 2.8	65.8 ± 2.6	67.4 ± 1.4
	CNN Baseline	66.8 ± 1.5	65.2 ± 0.9	68.2 ± 2.5	69.8 ± 1.1
CT-2 vs CT-0 (Moderate)	PAR-VAE (LogReg)	70.8 ± 1.2	66.5 ± 1.6	69.3 ± 2.0	74.6 ± 0.8
	CNN Baseline	68.3 ± 2.4	64.1 ± 2.1	66.7 ± 1.9	70.0 ± 1.5
CT-3 vs CT-0 (Severe)	PAR-VAE (RBF-SVM)	98.5 ± 1.1	97.3 ± 2.9	96.70 ± 3.79	99.3 ± 1.0
	CNN Baseline	57.1 ± 6.7	60.6 ± 6.3	57.22 ± 11.58	66.0 ± 7.5

criminative content. This is confirmed by Pearson correlation analysis (Fig. 2): 80% of learned latent dimensions have $R^2 > 0.7$ with linear or quadratic combinations of physics features, with only 20% capturing genuinely novel spatial information.

Table 4. Class overlap and separability (mean $\pm$ std, 3 seeds). Near- identical values for physics and learned features confirm the ceiling is data-intrinsic, not feature-design-dependent. A Cohen’s $d < 0.5$ across 60% of features means no classifier can reliably separate the classes using these features alone.

Task	Feature Type	Mean Overlap	Cohen’s d
CT-1 vs CT-0	Physics (14)	0.845 ± 0.080	0.38 ± 0.21
	Learned (top 15)	0.845 ± 0.070	0.41 ± 0.19
CT-2 vs CT-0	Physics (14)	0.841 ± 0.060	0.42 ± 0.15
	Learned (top 15)	0.841 ± 0.090	0.44 ± 0.18
CT-3 vs CT-0	Physics (14)	0.776 ± 0.050	0.42 ± 0.15
	Learned (top 15)	0.772 ± 0.050	0.368 ± 0.17

Fig. 1 shows PCA projections of the 85-dimensional latent space. Substantial class overlap with no decision boundary visible for CT-1 or CT-2 confirms the quantitative results visually. Physics attributes vary smoothly along the latent manifold, the signature of a biologically intrinsic ceiling rather than a representation failure.

5.4 Patch-Level Ablation

A natural hypothesis is that whole-lung averaging dilutes the focal GGO signal, and computing features at a finer spatial scale would improve discriminability. To test this directly, each CT slice was divided into a 3×3 spatial grid; all 14

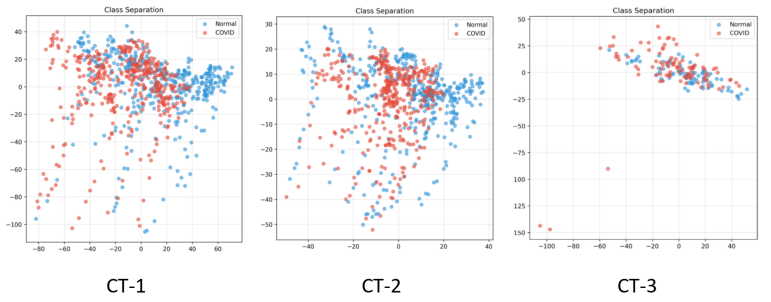


Fig. 1. PCA projections of the 85-dimensional latent space for CT-1 (left), CT-2 (middle) and CT-3(right). No separating boundary exists in either cohort; physics attributes vary smoothly along the manifold.

physics features were computed independently within each of the 9 patches (126 values per slice), and per-patch overlap coefficients were computed, as shown in Fig.3.

The result is counterintuitive but biologically consistent. Mean patch-level overlap (0.824) is *higher* than global overlap (0.765). Central patches (perihilar region) show the lowest overlap (≈ 0.800); lateral peripheral patches the highest (≈ 0.870) which is the inverse of what a spatial-averaging hypothesis would predict, since GGOs preferentially appear peripherally.

The biological explanation is direct. In mild disease, a peripheral GGO occupies a small fraction even of a peripheral patch; the feature is still diluted by the majority of normal tissue within that patch. Patches containing no GGO (the majority in CT-1) show artificially low HU values and sharper gradients than global statistics, because the weak averaging effect of a distant GGO has been removed. This increases within-class variance among COVID patches, thereby increasing overlap. The GGO detection problem, at the level of physically interpretable statistics, requires either manual lesion annotation or spatial pattern recognition beyond pixel-level statistics which is precisely the domain of black-box CNNs. The interpretability ceiling is biological and architectural, not a matter of aggregation scale.

5.5 Cross-Scanner Transfer: PAR-VAE Makes Domain Shift Visible

A key question for clinical deployment is how physics-aligned models behave across scanners. We evaluated the trained PAR-VAE (frozen encoder) on COVID-CT-MD [19], a second publicly available COVID-19 CT dataset acquired with a different scanner protocol. Physics alignment R^2 drops from 0.972 (MosMedData) to 0.320 (COVID-CT-MD); classification AUC on the new dataset is 0.71.

These numbers are not a model failure, but a diagnostic signal. The HU distribution difference between datasets is substantial: MosMedData mean HU within mask = +332 HU versus COVID-CT-MD mean HU = -150 HU ($\Delta = 482$

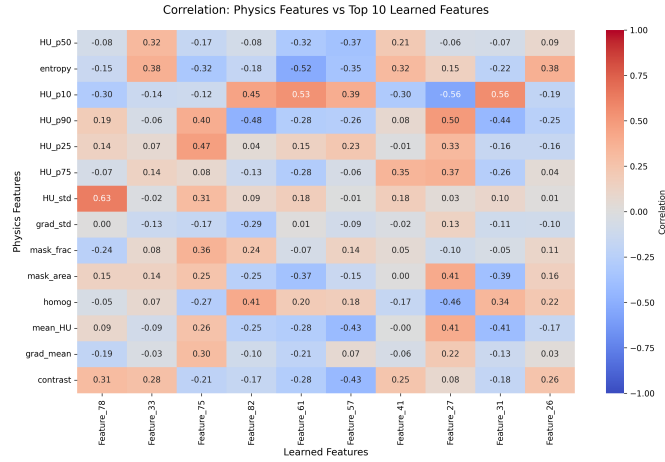


Fig. 2. Correlation between 14 physics attributes and top 10 learned latent dimensions. Most learned dimensions correlate strongly with physics features, confirming 80% representational redundancy between constrained and free portions of the latent space.

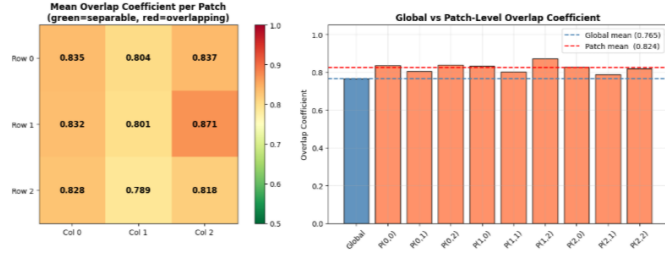


Fig. 3. Patch-wise overlap analysis reveals that the patch-wise feature overlap is in fact higher than the global feature overlap, due to increased in-class variance and inherent GGO structure

HU units). Mask area drops by 40% (144,964 vs. 84,742 pixels), indicating a different scanner field-of-view protocol. PAR-VAE’s R^2 drop to 0.320 directly quantifies this calibration gap, identifying *which features* degrade most and *how much* recalibration is needed before deployment.

The CNN baseline on the same transfer task achieves $AUC = 0.71$ with no equivalent internal signal that anything has changed. A clinician deploying CNN would receive confident predictions on miscalibrated data with no warning. PAR-VAE’s physics alignment score provides an explicit, quantitative deployment readiness indicator, a form of interpretability that extends beyond model internals to deployment diagnostics.

Table 5. Transfer learning performance on the COVID-MD dataset. Models trained on MosMedData (CT-3) exhibit a notable performance drop when evaluated on COVID-MD, highlighting the impact of scanner-induced domain shift, as captured by the PAR-VAE framework.

Setup	R^2	AUC	FN Rate
MosMedData (CT-3, in-domain)	0.972	0.999	1.1%
COVID-MD (trained from scratch)	0.322	0.710	49.3%
Frozen encoder + fine-tuned predictor	0.417	0.745	33.0%

6 Ablation Studies

Table 6 confirms the 3-phase annealing curriculum is necessary. Constant weights cause posterior collapse regardless of their values. Premature high λ causes attribute lock-in before reconstruction stabilizes, producing representations that predict physics attributes but cannot encode image content faithfully. Only the 3-phase schedule achieves healthy KL divergence ($KL \approx 15$) with stable physics alignment across all seeds.

Table 6. Annealing strategy ablation. KL divergence is the primary training health diagnostic. Posterior collapse ($KL < 5$) renders the latent space uninformative for downstream classification.

Strategy	KL at Epoch 50	Outcome
No annealing	< 5 (collapse)	Degenerate latent space
High β early	< 5 (collapse)	No image structure encoded
High λ early	≈ 15	Attribute lock-in, poor reconstruction
3-phase (ours)	≈ 15	Stable, generalizable alignment

The 85-dimensional latent space was validated against 64 and 96 dimensions. The 85-dimensional configuration achieves the smallest val-test R^2 gap (0.003), indicating optimal generalization. At 96 dimensions the gap widens to 0.056, indicating overfitting; at 64 dimensions the model lacks capacity to simultaneously encode all 14 physics attributes alongside residual spatial information.

7 Discussion and Conclusion

7.1 The Biological Ceiling and Its Clinical Consequences

The model-independent confirmation that both PAR-VAE and CNN hit the same $\approx 67\%$ AUC ceiling on mild COVID-19 is the central finding. This ceiling is not a PAR-VAE limitation but rather a property of the data. Early GGOs

produce changes in whole-lung statistics that are smaller than normal inter-patient variation, and no amount of architectural sophistication changes this if the representation remains global. The patch ablation proves this cannot be resolved by spatial refinement alone.

What separates the models is their behavior at scale. As disease severity increases and the ceiling lifts, global statistics finally shift enough to discriminate and PAR-VAE benefits fully ($99.3 \pm 1.0\%$ AUC at CT-3) while CNN, having learned shortcuts, does not, thereby remaining at $66.0 \pm 7.5\%$ AUC, missing $46.9 \pm 19.7\%$ of severe cases (ranging from 27.6% to 66.9% across seeds, compared with PAR-VAE’s consistent 1.6% miss rate). The mean 46.9% miss rate and the $\pm 19.7\%$ variance across seeds represent an clinical risk for any deployment without severity stratification.

For deployment, these findings suggest three design principles. First, interpretable systems should communicate explicit uncertainty for mild disease rather than applying undifferentiated confidence across all severities. Second, severity-stratified evaluation should be mandatory for COVID-19 CT AI systems since aggregate accuracy numbers conceal the severity gradient. Third, physics alignment scores provide a deployment readiness indicator across scanners that black-box models cannot offer.

7.2 Limitations and Future Work

Three limitations apply. First, MosMedData is a single cohort from a specific clinical context; the severity gradient magnitude may vary with scanner protocol and patient demographics. The COVID-CT-MD transfer experiment suggests this, though with results that are directional rather than definitive. Second, slices are treated as independent 2D samples; 3D volumetric context would allow confirmation of GGO persistence across slices, one of the key radiological cues our model cannot access.

In summary, we have shown that the performance ceiling of interpretable AI in early COVID-19 CT classification is biological, model-independent, and mechanistically explained. PAR-VAE achieves $R^2 = 0.972$ physics alignment, trails CNN narrowly on mild disease (67.4% vs 69.8% AUC, a 2.4pp gap), leads on moderate disease (74.6% vs 70.0% AUC), and critically avoids CNN’s catastrophic failure on severe disease ($99.3 \pm 1.0\%$ vs $66.0 \pm 7.5\%$ AUC; 1.6% vs $46.9 \pm 19.7\%$ miss rate). Physics constraints are not a performance compromise, but a clinical safety mechanism.

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